



What is Factor Analysis Transcript

Hello everyone, welcome to lesson 1 video 1. In this session, we will answer the question what exactly is factor analysis. Factor analysis is a statistical method that allows us to explore hidden structure behind large set of variables. In this module of dimensionality reduction, we are going to go through two important methods, one factor analysis and then we will also study about principal component analysis or which is called as PCA.

Both of these methods try to uncover hidden pattern in the data and we are first going to learn about factor analysis and then we will move into PCA. So what if you have these many information? Let's say you have these many texts. Are you happy with the service? Are you satisfied? Was your service joyful or did we meet your expectations? Now the common theme around all of these is service, right? The latent factor which is hidden within it or latent subject which is hidden within it is about service, the quality of service, right? Now these questions are not entirely independent.

If you think of the first is very correlated with second, right? All of these are somewhat correlated with each other. So the entire idea of factor analysis is to find which particular information are similar in terms of the latent factor and what they actually are talking about, right? And once we know what factors are actually correlated, we club them together as a factor. That's the entire idea about or high level idea behind it but we will get into the details of each and everything that we need to know from factor analysis in a bit, okay? First thing that you need to understand is factor analysis.

It uncovers latent factors and these latent factors are derived from the initial given data that you have, okay? Let's say you have a lot of variables which are correlated. So it tries to find you some of those latent factors which are correlated and give you the information which becomes uncorrelated at the end. So it tries to find you factors which comes out of highly correlated features.

So let's say there are information on certain common ground, service, service quality, happiness of the customer and so on because these are almost going to be common things, right? So it clubs them together into one particular feature which is or one particular variable which are called as factors. This is the simplest example that we can think of. Let's say you have variable 1 which could be anything and let's say you have age of the individual, experience of the individual.

Now both of them could be highly related to each other, right? Highly correlated. So we have a latent factor which can actually cover both of them. Why do we need to have both of them when you have a factor which can actually cover both of the information that we have in these two columns, right? That's the essence of factor analysis.

Now because of this if you think of you have in your input data you have let's say these five variables, okay? Now because you are clubbing together similar variables or



correlated variables which talk about the same subject, then you get to only two factors because the last three are common subject items which are factor 2. The first two are common subject items which are factor 1. Now you do not have to work on or you do not have to work with these five factors or these five variables. You can work with these two newly discovered factors which could be used as variables, right? So that's the entire idea behind using factor analysis. I hope this was clear.

The key terminologies which are involved around this as observed value, okay? These are the actual variables we have in our data set that can be measured directly, okay? So let's say the answer to the survey question that we have, right? Then we have latent variables. These are unobserved underlying constructs that we believe are causing the observed variables to correlate. So what subject or what ground or what measure is making them correlated? That's what it is.

So let's say whether it is about customer satisfaction or it's about brand loyalty or xyz, right? Similarly, we have factor loadings. Factor loading is nothing but the coefficients. These are the coefficients of the latent variables.

Now, they help us around understanding whether the correlation between an observed variable and a factor, the factor and the observed variable, what is that correlation, right? A high loading suggests that the variable is strong and indicator of the underlying factor. That's what it means. And if you see, look at this equation, you have these loadings along with the factor.

You have factor which has been discovered out of many variables and these loadings are there. These loadings termed as l_{ji} , l_{j1} , l_{j2} and so on. These factor loadings with j , it means that, let's say for example, if you have j variables and you are working with first variable, so the first variable's entire equation would become $l_{11} f_1$ plus $l_{12} f_2$ and so on.

And this e is the unique error part which is not being explained by the factors. Obviously, all the data that you have, let's say the data which is stored in all the variables that we have, we cannot create factors for all of them. So everything which cannot be explained using the factors comes under e_j , which we call as error term.

Now, this is the entire idea behind vector analysis. I hope this was clear. The main thing that you have to know is if you have many variables, you are finding communalities or common things between them.

Once you find communalities, you can actually create those factors as communalities and rather than using the raw variables or the raw data that you have, you use these factors. In summary, factor analysis is all about finding patterns among variables and reducing complexity. It provides us with factors that represent the underlying dimensions of our data.



In the next video, we'll study the conditions needed to apply factor analysis on any dataset

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